WNDRVR
SUPPORT NETWORK

VXWORKS 7 DNS PROGRAMMER'S GUIDE, SR0650

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VxWorks 7 DNS Programmer's Guide, SR0650

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1. OVERVIEW

1. DNS for VxWorks

The VxWorks® 7 DNS Programmer's Guide describes the VxWorks domain name server (DNS), a distributed database that applications use to translate host names to IP addresses and vice versa.

DNS uses a client/server architecture. The client is called the *resolver*. The server may also be called the *name server*.

DNS for VxWorks supports the dynamic DNS network service that allows a DNS client to notify a DNS name server to change the active DNS configuration of its configured host names, addresses, or other information.

With dynamic DNS, a computer can automatically associate its current IP address with a domain name. As a result, a remote user can resolve the host name used for the dynamic DNS service entry to the current address of the computer with a DNS query.

Source Code Files

The source code files for the DNS client are in the following directory:

installDir **/vxworks-7/pkgs_v2/net/ipnet/app/dnsc-***x.x.x.x***/**

Next topic: Supported RFCs for DNS on page 1

2. Supported RFCs for DNS

Domain name server (DNS) request for comments (RFCs) contain standards and design information defined by the Internet Engineering Task Force (IETF).

DNS for VxWorks 7 complies with the following IETF RFCs:

- RFC 1034: *Domain Names OE Concepts and Facilities*
- RFC 1035: *Domain Names OE Implementation and Specification*
- RFC 1886: *DNS Extensions to support IP version 6*
- RFC 2136: *Dynamic Updates in the Domain Name System (DNS UPDATE)*
- RFC 2553: *Basic Socket Interface Extensions for IPv6*
- RFC 2845: *Secret Key Transaction Authentication for DNS (TSIG)*
- RFC 3152: *Delegation of IP6.ARPA*
- RFC 3596: *DNS Extensions to Support IP Version 6*
- RFC 3646: *DNS Configuration options for Dynamic Host Configuration Protocol for IPv6*

The Wind River DNS implementation (IPDNSC) has the following exceptions to the DNS standards:

- DNS for VxWorks 7 does not allow iterative resolver lookups. The DNS client requires the DNS server to enable recursive lookups.
- DNS for VxWorks 7 does not allow A6 records.

Previous topic: DNS for VxWorks on page 1
Next topic: About DNS Technology on page 2

3. About DNS Technology

The domain name server (DNS) is comprised of three major components: domain namespace and resource records, name servers, and resolvers.

Domain Name Space

The naming system that DNS uses is a hierarchical and logical tree structure called the *domain namespace*. The namespace defines the structure of DNS names and provides the rules for their creation.

DNS names are DNS domains, computers, and services. For example, the Internet uses some of its domain names to identify hosts.

Name servers

Name servers hold information about the domain. In general, a particular name server has complete information about a subset of the domain space, and pointers to other name servers used to find information from any other part of the domain tree. Name servers that have complete information about a domain tree are said to be an authority for that part of the namespace. Authoritative information is organized into units called *zones*, and these zones are automatically distributed to the name servers, which provide redundant services for the data in a zone.

Resolvers

Resolvers, or DNS clients, extract information from name servers in response to client requests. Resolvers must be able to access at least one name server and use that name server's information to answer a query either directly, or by pursuing the query using referrals to other name servers. A resolver is typically a system function that is directly accessible to your program. Therefore, no protocol is necessary between the resolver and your program.

Previous topic: Supported RFCs for DNS on page 1
Next topic: DNS Features and Functions on page 2

4. DNS Features and Functions

The domain name server (DNS) client for VxWorks 7 allows name-to-address lookups as well as address-to-name lookups for systems running IPv4 or dual IPv4/IPv6 stacks.

For a list of implemented request for comments (RFCs) and exceptions, see DNS Operational Requirements on page 10.

The DNS client includes the following features:

- A DNS client (stub resolver) conforming to RFC 1034 and RFC 1035.
- One primary and up to three backup name servers.
- Local caching of resource records.
- Name-to-address lookups capable of querying for IPv4 addresses as well as IPv6 (RFC 3596) addresses.
- Address-to-name lookups for IPv4 and IPv6 (RFC 3596) addresses.

  The DNS client includes the following functions:
- The ipdnsc_getipnodebyname( ) and ipdnsc_getipnodebyaddr( ) functions as per RFC 2553. These functions are reentrant versions of the gethostbyname( ) and gethostbyaddr( ) functions. Use them to perform DNS lookup of both IPv4 and IPv6 addresses.
- As part of dynamic DNS, the ipdnsc_do_update( ) function allows a client to update name servers with client information. This function handles a complete DNS update process. It creates a dynamic update packet for each zone and sends it to the name

servers for that zone. It then receives and parses the response. For example usage, see the ipdnsc_test_update( ) function in **ipdnsc_api.c**.

- The ipdnsc_freehostent( ) function frees memory allocated for an ip_hostent( ) structure.
- The ipdnsc_cache_flush( ) function removes all entries in the resolver cache.
- A shell command, nslookup to perform a name-to-address or address-to-name lookup.

Previous topic: About DNS Technology on page 2

2. DNS CONFIGURATION AND BUILD

1. DNS Configuration and Build Workflow

The VxWorks 7 domain name server (DNS) installation includes source code but does not include pre-built kernel libraries.

Before beginning any VxWorks development, you must build the VxWorks kernel libraries (a VxWorks source build (VSB) project). You must also build the VxWorks run-time kernel image based on the kernel libraries (a VxWorks image project (VIP)).

For specific instructions on how to configure and build your VSB and your VIP, see the VxWorks 7 Configuration and Build Guide.

For a list of available source build configuration options for DNS, see DNS Source Build Configuration Options (VSB) on page 4.

For a list of available kernel configuration components and parameters for DNS, see DNS Kernel Configuration Components (VIP) on page 4.

Next topic: DNS Source Build Configuration Options (VSB) on page 4

2. DNS Source Build Configuration Options (VSB)

Domain name server (DNS) configuration includes VxWorks source build (VSB) options.

To include DNS support in your VSB, include the following option:

Option	Description
IPNET_DNSC	Include DNS client in the VSB.

Previous topic: DNS Configuration and Build Workflow on page 4
Next topic: DNS Kernel Configuration Components (VIP) on page 4

3. DNS Kernel Configuration Components (VIP)

Domain name server (DNS) configuration includes VxWorks image project (VIP) kernel configuration components.

To include support for the DNS client in your kernel image (VIP project), include the DNS Client (INCLUDE_IPDNSC) component.

> Note: When your VxWorks image project (VIP) is based on the IPv4 libraries only, IPv6 queries are not supported.

Previous topic: DNS Source Build Configuration Options (VSB) on page 4
Next topic: DNS Optional Kernel Configuration Components on page 4

4. DNS Optional Kernel Configuration Components

You can modify the domain name server (DNS) client behavior with a number of optional shell command components.

Component	Description
INCLUDE_IPNSLOOKUP_CMD	Include the ability to query Internet name servers interactively (**nslookup**).
INCLUDE_IPCOM_SYSVAR_CMD	Include the system variables shell command (**sysvar**).
INCLUDE_IPD_CMD	Include the daemon control shell command (**ipd**).

Previous topic: DNS Kernel Configuration Components (VIP) on page 4
Next topic: DNS Client Configuration Parameters on page 5

5. DNS Client Configuration Parameters

You can modify the Wind River domain name server (DNS) client behavior with a number of configuration parameters and system variables.

Parameter Name, System Variable Name, and Description	Default and Type
DNSC_CFGFILE The full path of the configuration file that VxWorks will use to configure the DNS client parameters. When no file is specified (default setting), the DNS client parameter values are based on the VIP configuration. For more information, see About DNS File-Based Configuration on page 7.	" " char *
DNSC_DOMAIN_NAME **ipdnsc.domainname** The domain where the local host is located. If you do not define this parameter, DNS is not aware of the local domain name and DNS users must provide fully qualified domain names in address lookups even for hosts in the same domain as the local host.	**"windriver.com"** char *
DNSC_PRIMARY_NAME_SERVER **ipdnsc.primaryns** The address of the primary name server. If you do not define this parameter, DNS does not use any primary name server.	" " char *
DNSC_SECONDARY_NAME_SERVER **ipdnsc.secondaryns** The address of the secondary name server. If you do not define this parameter, DNS does not use a secondary name server.	" " char *
DNSC_TERTIARY_NAME_SERVER **ipdnsc.tertiaryns**	" " char *

Parameter Name, System Variable Name, and Description	Default and Type
The address of the tertiary name server. If you do not define this parameter, DNS does not use a tertiary name server.	
DNSC_QUATERNARY_NAME_SERVER **ipdnsc.quaternaryns** The address of the quaternary name server. If you do not define this parameter, DNS does not use a quaternary name server.	" " char *
DNSC_SERVER_PORT **ipdnsc.port** The port that the DNS client uses for DNS queries. If you do not define this parameter, the DNS client uses the default DNS port.	**"53"** char *
DNSC_RETRIES **ipdnsc.retries** The number of retries on each name server. If you do not define this parameter, DNS uses the default value.	**"2"** char *
DNSC_TIMEOUT **ipdnsc.timeout** The number of seconds before a retry if the DNS server fails to answer. If you do not define this parameter, DNS uses the default value.	**"10"** char *
DNSC_IP4_ZONE **ipdnsc.ip4.zone** The zone for the DNS client to use for address-to-name lookups of IPv4 addresses.	**"in-addr.arpa"** char *
DNSC_IP6_ZONE **ipdnsc.ip6.zone** The zone for the DNS client to use for address-to-name lookups of IPv6 addresses. Use the string ip6.int for RFC 1886 or ip6.arpa for RFC 3152.	**"ip6.int"** char *

Previous topic: DNS Optional Kernel Configuration Components on page 4

3. FILE-BASED CONFIGURATION

1. About DNS File-Based Configuration

By default, domain name server (DNS) client parameter values are set at VxWorks image project (VIP) configuration time. The file-based configuration functionality allows you to reconfigure DNS client parameter values at boot time.

Configure DNS client parameter values with a list of system variables stored in a text-based configuration file. The file path and file name are specified by a VIP parameter. This configuration method provides an easy way to adapt a system to new networking environments by simply updating the configuration file contents and then transferring the file to the target before starting the DNS client.

Next topic: Enabling and Disabling DNS File-Based Configuration on page 7

2. Enabling and Disabling DNS File-Based Configuration

Configure the domain name server (DNS) client from a configuration file or in a VxWorks image project (VIP).

About This Task

To enable the configuration of the DNS client from a configuration file, specify the full path to the configuration file as part of the DNSC_CFGFILE VIP parameter.

To disable the configuration of the DNS client from a configuration file, set the DNSC_CFGFILE VIP parameter to NULL (**""**). The DNS client configuration is then based on the VIP configuration.

For information on configuring the **DNSC_CFGFILE** VIP parameter, see DNS Client Configuration Parameters on page 5.

For information on operational requirements for the target, see DNS Operational Requirements on page 10.

Previous topic: About DNS File-Based Configuration on page 7
Next topic: DNS Configuration File Format on page 7

3. DNS Configuration File Format

The domain name server (DNS) configuration file is a text-based file with a specific format. Edit the file with a standard text editor.

To configure a parameter from the configuration file, a valid system variable name must be followed by a valid value (system variable / value pair) and be on a single line.

A configuration file can also contain comments.

> Note: The configuration file does not accept escape characters (\).

Valid system variables are provided in the sample configuration file.

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A system variable/value pair can start in any column of a given line. The system variable and name must be separated by least one space or tab.

Each line can only contain a single system variable/value pair. Boolean, integer, floating point, and string values can be listed with or without double quotation marks.

Example valid line content follows:

```
systemVariableName_A  TRUE
systemVariableName_B  2
systemVariableName_C  192.168.1.1
systemVariableName_D  String
systemVariableName_A  "TRUE"
systemVariableName_B  "2"
systemVariableName_C  "192.168.1.1"
systemVariableName_D  "String"
```

For example:

```
# This is a comment line.
```

Add a comment on the same line as a system variable/name pair. For example:

```
systemVariableName_A  FALSE          # This is a boolean value.
systemVariableName_B  22             # This is an integer value.
systemVariableName_C  192.168.2.2    # This is a floating point value.
systemVariableName_D  String         # This is a string value.
```

Previous topic: Enabling and Disabling DNS File-Based Configuration on page 7
Next topic: DNS Configuration File Processing on page 8

4. DNS Configuration File Processing

The domain name server (DNS) file processing differs based upon the content of the file. All empty lines and comment lines are not processed.

If a configuration file is not found, no processing is done and the system variables retain their values from the VIP configuration.

If a configuration file is located, the system processes the configuration file line-by-line. The values in the configuration file overwrite the values set at VIP configuration time.

Lines with errors such as misspelled system variable names and missing parameter values are ignored (skipped). Hence, a system variable associated with a faulty line entry retains the value set at VIP configuration time.

Note: If a system variable/value pair is listed more than once in a file, the last valid occurrence is used to configure the system variable. No error is reported as part of the diagnostic output.

Previous topic: DNS Configuration File Format on page 7
Next topic: Sample DNS Client Configuration File on page 9

5. Sample DNS Client Configuration File

The sample domain name server (DNS) client configuration file contains a list of valid system variables that can be modified to change the DNS client behavior.

The sample configuration file is located at:

installDir **/vxworks-7/pkgs_v2/net/ipnet/app/dnsc/sample_cfg/etc/ dnsc.conf**

Note: Only the specified system variables in the sample file are configurable from a configuration file.

To correlate DNS client VIP parameter names to DNSC client system variables listed in the sample configuration file, see DNS Client Configuration Parameters on page 5.

Note: To prevent unnecessary reconfiguration of the VIP parameters, the configuration file should only contain system variables that must be changed. The remaining parameter values will be taken from the VIP configuration.

Previous topic: DNS Configuration File Processing on page 8
Next topic: DNS Diagnostics on page 9

6. DNS Diagnostics

You can enable and disable domain name server (DNS) diagnostics. When enabled, the diagnostics process generates output and provides status on the action performed on each system variable.

Enable and disable diagnostics from within the configuration file. When enabled, the configuration status of the system variables is sent to the console, if a console is available.

To enable diagnostics, set the *cfgfile.diag* variable in the configuration file to **TRUE**. To disable, set to **FALSE**. For example:

```
cfgfile.diags      FALSE
```

An updated system variable is labeled as **Updated**. The output format is as follows:

```
Variable: systemVariableName Value: value Result: Updated
```

A skipped update for a system variable is labeled as **Unrecognized variable(Skipping)**. The output format is as follows:

```
Variable: systemVariableName Value: value Result: Unrecognized variable (Skipping)
```

Previous topic: Sample DNS Client Configuration File on page 9
Next topic: Verifying DNS System Variables on page 9

7. Verifying DNS System Variables

View the domain name server (DNS) system variables and their content with a shell command.

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About This Task

To view system variables, use the following command:

```
vxWorks# sysvar list serviceName
```

Note: To access the *sysvar* shell command, you must add the INCLUDE_IPCOM_SYSVAR_CMD component to your VIP.

Previous topic: DNS Diagnostics on page 9
Next topic: DNS Operational Requirements on page 10

8. DNS Operational Requirements

To enable parameter configuration from a file, the target hosting the domain name server (DNS) client must include read/write mass storage capability. The file is transferred to the host by one of several methods.

A configuration file can be modified by an external host and transferred to the target using one of the following methods:

- By transferring the file from an FTP server to the target.
- By mounting the target's mass storage device as a USB disk and using desktop system facilities to update the configuration file.
- By using a removable mass storage device supported by the target.

Previous topic: Verifying DNS System Variables on page 9

4. SHELL COMMANDS

1. About DNS Shell Commands

The domain name server (DNS) shell commands include *nslookup, sysvar, ipd* and *syslog.*

The shell commands perform the following functions:

- nslookup shell command to query Internet name servers interactively.
- sysvar shell command to modify DNS client configuration parameters.
- ipd shell command to view the state of the DNS client service.
- syslog shell command to set the priority for the DNS client syslog debug messages.

To run shell commands, you must access the command-interpreter mode. To access the command-interpreter mode from the kernel shell, enter the following command:

```
-> cmd
```

```
[vxWorks *} #
```

To return to the kernel shell, enter the following command:

```
# C
```

```
->
```

Next topic: Using the Name Server Lookup Shell Command on page 11

2. Using the Name Server Lookup Shell Command

The nslookup command queries an Internet name server for information about a host.

About This Task

The *host* option may be a domain name or an Internet IPv4 or IPv6 address. The *nameServer* is optional; if you include it (as an Internet address), nslookup uses that name server rather than the default primary name server.

To query multiple name servers, query each server independently with a separate nslookup command.

The nslookup command syntax is as follows:

```
nslookup [-f] [-c] [-v version] [-t timeout] [-r retries] host [nameServer]
```

To get help for the nslookup command, use the following command:

```
# nslookup
```


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Option	Description
-c	Use the domain name server (DNS) resolver's local cache. The default is to always perform the name server lookup and not to use the cache.
-f	Flush the DNS resolver cache and then exit immediately.
-r *retries*	Set the number of retries nslookup attempts if the name server fails to answer.
-t *timeout*	Set how long nslookup waits for an answer from the name server before it retries.
-v *version*	Set the type of address that is requested in a name-to-address lookup, either 4 (IPv4) or 6 (IPv6). If you do not indicate the version by using the -v flag, nslookup first tries to find an IPv6 address and then, if it cannot find one, it tries to find an IPv4 address.

Previous topic: About DNS Shell Commands on page 11
Next topic: Setting the DNS Client System Variables on page 12

3. Setting the DNS Client System Variables

You can view all domain name server (DNS) client sysvar variables from the kernel shell in command-interpreter mode.

About This Task

To view all DNS client sysvar variables, access the command-interpreter mode from the kernel shell and execute the following command:

```
# sysvar list ipdnsc
```

To change DNS client parameters with sysvar, use the sysvar set command.

Previous topic: Using the Name Server Lookup Shell Command on page 11
Next topic: Viewing the State of the DNS Client Service on page 12

4. Viewing the State of the DNS Client Service

Use the ipd command to view the state of the domain name server (DNS) client service (ipdnsc).

About This Task

For example:

```
# ipd list
```

```
Services:
...
ipdnsc              started
```

Previous topic: Setting the DNS Client System Variables on page 12
Next topic: Setting the DNS Client System Log Priority on page 13

5. Setting the DNS Client System Log Priority

Use the syslog command to obtain and change the priority level for domain name server (DNS) client logging. Run the syslog list command to get the current priority level for all modules.

About This Task

```
# syslog list
    syslog facility  priority
    ...
    ipdnsc Error
```

To list the available syslog priority levels, use the following command:

```
# syslog priority *
syslog: missing <prio>
        priorities: Emerg Crit Error Warning Notice Info Debug Debug2
```

Use the following command syntax to change the priority level for the DNS client:

```
# syslog priority ipdnsc priorityLevel
```

For more information on the syslog shell command, see the VxWorks 7 Network Stack Programmer's Guide.

Previous topic: Viewing the State of the DNS Client Service on page 12

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5. API FUNCTIONS

1. DNS Client API Functions

A number of API functions are available to the domain name server (DNS) client.

The DNS client public API functions are defined in **ipdnsc_api.c** and **ipdnsc.c**. The table below lists the API functions available to DNS client applications and gives a brief description of each function. For more information, see the reference entries for the DNS client (ipdnsc).

Function	Description
ipdnsc_getipnodebyname( )	Get host information by node name.
ipdnsc_getipnodebyaddr( )	Get host information by IP address.
ipdnsc_freehostent( )	Free memory allocated for an **ip_hostent** structure.
ipdnsc_cache_flush( )	Remove all entries in the resolver cache.
ipdnsc_do_update( )	Create and send a DNS update message and process the response.